A Comprehensive Parasitic-Aware Post-Layout Design and Verification Methodology for a 6.5-GHz Ultra-Low-Power Ring Oscillator in 65-nm CMOS

***Abstract*—This paper presents a comprehensive parasitic-aware post-layout design and verification methodology for a low-power three-stage CMOS ring oscillator implemented in a 65-nm Low-Power (LP) CMOS process. Operating from a 1.0-V supply, the oscillator is physically implemented using custom layout techniques with a compact active area of 73.96 μm^2. Unlike conventional schematic-level evaluations, the proposed framework incorporates detailed back-end-of-line (BEOL) parasitic extraction to assess the impact of interconnect effects on high-frequency operation. Following successful design rule checking (DRC), layout-versus-schematic (LVS) verification, and detailed standard parasitic format (DSPF) extraction, post-layout periodic steady-state (PSS) and periodic noise (`pnoise`) analyses are performed to characterize the oscillator under realistic conditions. The extracted design achieves a stable oscillation frequency of 6.50 GHz while consuming only 286.3 μW from a 1.0-V supply. The simulated single-sideband phase noise is -77.6 dBc/Hz at a 1-MHz offset, corresponding to a figure of merit (FoM) of -159.3 dBc/Hz. The comparison between schematic and post-layout simulations highlights the significant influence of distributed interconnect parasitics on oscillator performance, emphasizing the importance of parasitic-aware physical verification in nanoscale CMOS circuits. The proposed methodology provides a practical framework for the accurate design and evaluation of low-power, high-frequency oscillator blocks in advanced CMOS technologies.**

***Index Terms*—Ring oscillator, CMOS, post-layout verification, parasitic extraction, DSPF, phase noise, low-power design.**

I. Introduction

Oscillators are essential building blocks in modern integrated circuits and are widely used in phase-locked loops (PLLs), clock-and-data recovery (CDR) circuits, and other timing systems [4]. Among various oscillator topologies, ring oscillators are attractive because they occupy a small silicon area, do not require inductors, and can be readily implemented in standard CMOS processes [1], [3]. In particular, the three-stage ring oscillator offers a simple structure and naturally provides multiple phase outputs, making it suitable for many high-speed applications [5], [10].

As CMOS technology scales into the nanometer regime, however, maintaining the accuracy of circuit performance prediction becomes increasingly challenging. During the early design stage, circuit performance is commonly evaluated using schematic-level simulations. While this approach is useful for initial design exploration, it does not account for parasitic elements introduced during physical implementation. At multi-gigahertz frequencies, interconnect resistances and capacitances in the back-end-of-line (BEOL) layers can no longer be neglected. These parasitic components increase signal propagation delay and may cause the actual oscillation frequency to deviate from the intended design value [7].

For this reason, post-layout verification plays an important role in high-frequency circuit design. Although many studies have reported the performance of CMOS ring oscillators, most discussions focus on transistor-level simulations, while the influence of layout parasitics is often treated only briefly. As technology nodes continue to shrink, understanding the difference between schematic and post-layout performance becomes increasingly important for obtaining reliable circuit behavior.

In this work, a low-power three-stage ring oscillator is implemented in a 65-nm Low-Power (LP) CMOS process and evaluated using a complete post-layout design flow. After layout generation, the circuit is verified through design rule checking (DRC) and layout-versus-schematic (LVS) procedures, followed by RC parasitic extraction in Detailed Standard Parasitic Format (DSPF). The extracted netlist is then analyzed using periodic steady-state (PSS) and periodic noise (`pnoise`) simulations to characterize the oscillator under realistic operating conditions.

The main contributions of this paper are summarized as follows:

- A complete physical design and post-layout verification flow for a three-stage CMOS ring oscillator in a 65-nm LP CMOS technology.
- A comparison between schematic-level and extracted-netlist simulations to evaluate the effect of BEOL interconnect parasitics on oscillation frequency and propagation delay.
- Post-layout characterization of a compact 6.5-GHz ring oscillator with a power consumption of 286.3 μW, including phase-noise and figure-of-merit analysis.

II. Ring Oscillator Operation and Design Methodology

A. Oscillation Principle and Startup Condition

The proposed oscillator consists of three cascaded CMOS inverter stages connected in a closed loop. Each stage is formed by a PMOS–NMOS pair, and the output of the last stage is fed back to the input of the first stage. Because the loop contains an odd number of inversions, the circuit is able to generate a self-sustained periodic signal.

To maintain oscillation, the loop must satisfy the Barkhausen criterion at the oscillation frequency. This requires the total phase shift around the loop to reach 360° and the loop gain to be equal to or greater than unity [1]. In a three-stage ring oscillator, the inverter chain contributes an overall phase inversion of 180°, while the frequency-dependent phase shift provided by the stages contributes the remaining 180° required for sustained oscillation.

Assuming that all stages are identical, the oscillation frequency can be related to the dominant pole frequency of each stage as

$$\omega_0 = \omega_p \tan\left(\frac{\pi}{N}\right) \tag{1}$$

where N is the number of stages and ω_p is the dominant pole frequency of a single stage. For the three-stage configuration used in this work ($N = 3$), the oscillation frequency becomes

$$\omega_0 = \sqrt{3}\,\omega_p \tag{2}$$

Although (2) provides a useful analytical relationship, the oscillation frequency is more conveniently expressed in terms of the propagation delay of each stage. For an N-stage ring oscillator, the oscillation frequency can be approximated by

$$f_{\text{osc}} = \frac{1}{2N\tau_p} \tag{3}$$

where τ_p denotes the average propagation delay of one stage. Since the propagation delay is mainly determined by the effective resistance and load capacitance of the inverter, it can be approximated by

$$\tau_p \approx 0.69\,R_{\text{eq}}C_{\text{tot}} \tag{4}$$

where R_{eq} is the equivalent switching resistance and C_{tot} is the total capacitance seen at the output node. Substituting (4) into (3) gives

$$f_{\text{osc}} \approx \frac{1}{1.38\,NR_{\text{eq}}C_{\text{tot}}} \tag{5}$$

In practical implementations, the total capacitance consists of both intrinsic load capacitance and parasitic capacitance introduced during physical layout. Therefore,

$$C_{\text{tot}} = C_L + C_{\text{par}} \tag{6}$$

where C_L represents the intrinsic load capacitance and C_{par} denotes the parasitic capacitance associated with interconnect routing and device layout. Consequently, the oscillation frequency can be rewritten as

$$f_{\text{osc}} \approx \frac{1}{1.38\,NR_{\text{eq}}\,(C_L + C_{\text{par}})} \tag{7}$$

Equation (7) shows that additional parasitic capacitance increases the stage delay and reduces the oscillation frequency. This effect becomes increasingly significant at multi-gigahertz frequencies, where interconnect parasitics can no longer be neglected. Therefore, accurate prediction of oscillator performance requires post-layout parasitic extraction and verification, which form the main focus of this work.

B. Short-Channel Propagation Delay Model

Although the startup condition of a ring oscillator can be explained using small-signal analysis, its steady-state oscillation frequency is primarily determined by the propagation delay of the inverter stages. For an N-stage ring oscillator, one complete period corresponds to the time required for a signal transition to propagate twice around the loop. Therefore,

$$T = \frac{1}{f_{\text{osc}}} = 2Nt_d \tag{8}$$

where t_d denotes the average propagation delay of a single stage.

In long-channel devices, the propagation delay is commonly approximated by assuming a constant charging or discharging current driving a lumped node capacitance. Under this assumption,

$$t_d \approx \frac{C_{\text{node}}V_{\text{DD}}}{2I_{\text{drive}}} \tag{9}$$

where C_{node} represents the effective node capacitance and I_{drive} is the average switching current.

However, this approximation becomes less accurate in deep-submicron technologies. In 65-nm CMOS devices, velocity saturation and mobility degradation limit the available drain current. Under strong inversion, the saturation current can be approximated as

$$I_{\text{sat}} = v_{\text{sat}}C_{\text{ox}}W\,(V_{\text{GS}} - V_{\text{th}} - V_{\text{dsat}}) \tag{10}$$

where v_{sat} is the carrier saturation velocity, C_{ox} is the oxide capacitance per unit area, W is the effective transistor width, V_{GS} is the gate-to-source voltage, V_{th} is the threshold voltage, and V_{dsat} is the drain saturation voltage.

By considering the average delay during both high-to-low and low-to-high transitions, the stage delay can be approximated by

$$t_d = \frac{t_{\text{pHL}} + t_{\text{pLH}}}{2} \approx \frac{C_{\text{node}}V_{\text{DD}}}{2v_{\text{sat}}C_{\text{ox}}\left(\frac{W_n+W_p}{2}\right)(V_{\text{DD}} - V_{\text{th}} - V_{\text{dsat}})} \tag{11}$$

Equation (11) indicates that the oscillation frequency is strongly affected by the effective node capacitance. In schematic-level simulations, C_{node} mainly consists of intrinsic device capacitances. After physical implementation, however, additional interconnect capacitances and routing parasitics contribute to the total node capacitance. Consequently,

$$C_{\text{node}} = C_{\text{intrinsic}} + C_{\text{par}} \tag{12}$$

where C_{par} denotes the parasitic capacitance introduced by layout interconnections. An increase in C_{par} results in a larger propagation delay and therefore reduces the oscillation frequency. This effect is examined quantitatively through post-layout extraction in the following sections.

C. Pre-Layout Performance Baseline

Fig. 1 shows the schematic of the proposed three-stage CMOS ring oscillator. To provide a reference for evaluating the impact of layout parasitics, the circuit was first simulated at the schematic level using a supply voltage of $V_{\text{DD}} = 1.0$ V under typical-typical (TT) process conditions at 27°C. All transistors employ a channel length of 60 nm. The NMOS and PMOS devices are sized to $W_n = 2$ μm and $W_p = 4$ μm, respectively, to provide balanced drive strength.

TABLE I
TRANSISTOR DIMENSIONS OF THE THREE-STAGE RING OSCILLATOR

Device Class	Model	Width (W)	Length (L)
NMOS (3 devices)	`nch`	2.0 μm	60 nm
PMOS (3 devices)	`pch`	4.0 μm	60 nm

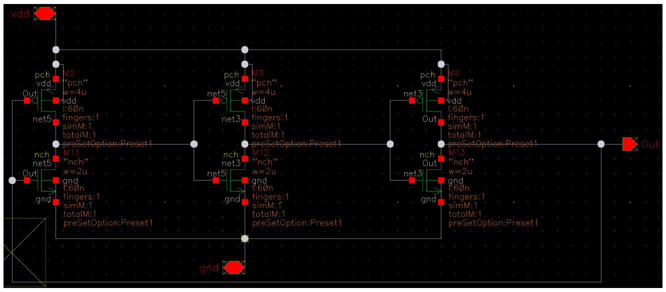

Fig. 1. Schematic capture of the verified three-stage static CMOS ring oscillator loop.

Under these conditions, the transient simulation produces an oscillation frequency of

$$f_{\text{osc,schematic}} = 11.15 \text{ GHz} \tag{13}$$

which corresponds to a period of

$$T_{\text{schematic}} = \frac{1}{f_{\text{osc,schematic}}} = 89.71 \text{ ps}. \tag{14}$$

Using (8), the average propagation delay of a single inverter stage can be estimated as

$$t_{\text{d,schematic}} = \frac{T_{\text{schematic}}}{2N} = 14.95 \text{ ps} \tag{15}$$

for the three-stage configuration ($N = 3$). These results serve as the pre-layout performance baseline and are later used to quantify the influence of interconnect parasitics extracted from the physical layout.

III. Physical Design and Custom Layout Methodology

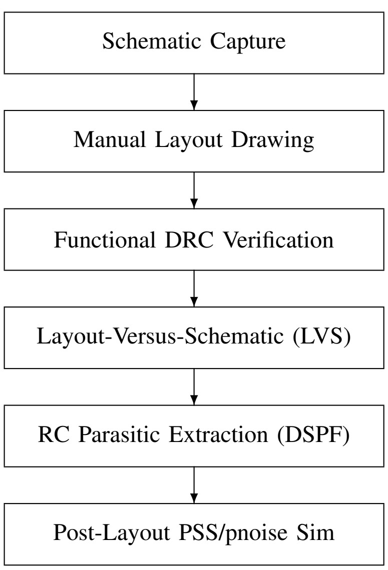

Fig. 2. Custom physical-design and parasitic-aware verification flow framework implemented in this work.

A. Device Sizing and Symmetry Considerations

The ring oscillator was implemented in a 65-nm Low-Power (LP) CMOS process. To obtain balanced rise and fall transitions, the PMOS devices were sized twice as wide as the NMOS devices, with $W_p = 4$ μm and $W_n = 2$ μm. This sizing ratio compensates for the lower hole mobility of PMOS transistors and helps maintain similar pull-up and pull-down drive strengths [2].

All transistors employ the minimum channel length of 60 nm to maximize switching speed and reduce intrinsic delay [1]. The resulting symmetric inverter structure improves transition balance and helps minimize duty-cycle distortion in the oscillation waveform.

B. Floorplanning and Interconnect Routing

A fully custom layout was used to implement the oscillator core. To simplify routing and maintain symmetry, the three PMOS devices were placed in the upper row within a common n-well, while the corresponding NMOS devices were arranged in the lower row. This arrangement provides a compact structure and reduces unnecessary routing overhead.

The feedback paths between stages were routed using short interconnects primarily in the lower metal layers. Since the oscillator operates in the multi-gigahertz range, minimizing interconnect length is important to reduce parasitic resistance and capacitance [2]. The final layout occupies an area of 8.6 μm $\times$ 8.6 μm, corresponding to a total core area of 73.96 μm^2.

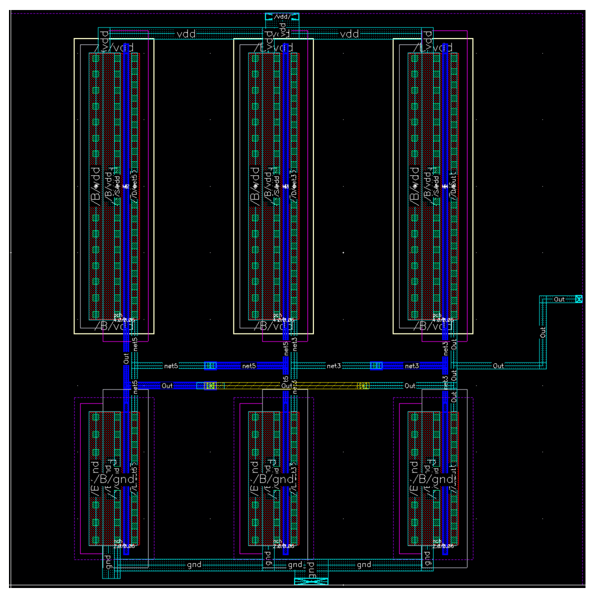

Fig. 3. Manually routed layout core showing row-aligned PMOS and NMOS device matching (8.6 μm $\times$ 8.6 μm).

C. Design Rule Checking

After layout generation, Design Rule Checking (DRC) was performed to verify compliance with the design rules of the 65-nm CMOS process. During the initial verification stage, a Metal-1 spacing violation (rule M1.S.1, min 0.09 μm) was identified and corrected by adjusting the routing geometry. After modification, the layout satisfied all essential DRC requirements. Some remaining messages were associated with density-related rules and chip-level layers (CSR and AP layers), which do not affect the functionality of the oscillator core and are routinely addressed via automated dummy-fill insertion during final mask assembly.

D. Layout-Versus-Schematic Verification

Layout-Versus-Schematic (LVS) verification was carried out to ensure consistency between the physical layout and the original schematic. The comparison includes device types, transistor dimensions, pin assignments, and net connectivity. The extracted layout was found to be consistent with the schematic, indicating that the designed topology was correctly transferred into the physical implementation.

Consequently, the effects observed in the post-layout simulations can be attributed mainly to the parasitic elements introduced by interconnects and device geometries rather than connectivity errors.

IV. Parasitic Extraction and Post-Layout Simulation Methodology

A. RC Parasitic Extraction

Following DRC and LVS verification, RC parasitic extraction was performed to generate a Detailed Standard Parasitic Format (DSPF) netlist containing distributed resistance and capacitance associated with interconnects and device geometries [2]. Compared with the schematic-level representation, the extracted netlist provides a more realistic description of the oscillator under practical operating conditions.

TABLE II
Extracted Per-Net Parasitic Capacitance from DSPF

Extracted Net Label	Parasitic Capacitance (fF)
`Out`	4.07
`net3`	3.54
`net5`	3.52
`Vdd`	3.16
`gnd`	3.01

Table II summarizes the parasitic capacitances associated with the major circuit nodes. The output node (`Out`) exhibits the largest capacitance of 4.07 fF because of the feedback routing and external observation port, while the internal nodes `net3` and `net5` show similar capacitance values of 3.54 fF and 3.52 fF owing to the symmetrical layout arrangement. The localized power rails (`Vdd` and `gnd`) contribute 3.16 fF and 3.01 fF to the substrate. These additional parasitic components increase the effective node loading and are expected to increase the propagation delay of each stage. Consequently, the oscillation frequency obtained from the extracted netlist is lower than that predicted by the ideal schematic model. The influence of these parasitic components is analyzed quantitatively in the following section.

B. Post-Layout Simulation Methodology

Post-layout characterization used periodic steady-state (PSS) and periodic noise (`pnoise`) analyses. Because the extracted DSPF netlist contains many distributed RC elements, the simulation settings were tuned for stable convergence and accurate steady-state solutions. The default `analogLib port` primitive presents a 50 Ω source/load resistance. Although this matches standard test instruments, it overloads an unbuffered, low-power sub-micron inverter stage, suppressing the loop gain below the Barkhausen criterion and extinguishing the oscillation. A high-impedance (1 MΩ) output port was therefore used to minimize core loading. The PSS analysis ran in oscillator mode with sufficient stabilization time for the amplitude and frequency to settle before the shooting algorithm was applied. For phase noise, `pnoise` was swept over offsets from 1 kHz to 100 MHz using the time-average method with phase-modulation (PM) noise as the output. These settings yielded stable simulations and reliable extraction of oscillation frequency, power consumption, and phase-noise characteristics.

C. Phase Noise and Figure-of-Merit Evaluation

The phase-noise performance of the oscillator was evaluated from the `pnoise` simulations [6], [9]. Single-sideband phase noise was measured at various offset frequencies relative to the carrier. To facilitate comparison with previously reported oscillators, the figure of merit (FoM) was calculated as [8]

$$\mathrm{FoM} = \mathcal{L}(\Delta f) - 20\log_{10}\left(\frac{f_0}{\Delta f}\right) + 10\log_{10}\left(\frac{P}{1\ \mathrm{mW}}\right) \quad (16)$$

where $\mathcal{L}(\Delta f)$ is the single-sideband phase noise at an offset frequency Δf, f_0 denotes the oscillation frequency, and P represents the power consumption.

V. Results and Discussion

A. Post-Layout Transient Characteristics

The post-layout transient response of the RC-extracted oscillator was characterized under a 1.0-V supply. Fig. 4 shows stable periodic oscillations with nearly rail-to-rail voltage swings ranging from approximately -0.05 V to 1.04 V.

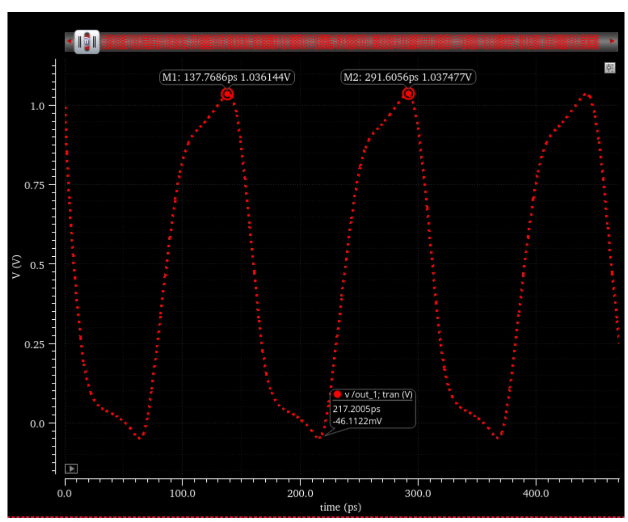

Fig. 4. Post-layout (RC-extracted) transient output waveform showing stable rail-to-rail oscillation at 6.50 GHz with a period of 153.84 ps.

From the steady-state waveform, the oscillation period was measured as

$$T_{\text{post-layout}} = 153.84\ \text{ps} \quad (17)$$

which corresponds to an oscillation frequency of

$$f_{\text{osc,post-layout}} = \frac{1}{T_{\text{post-layout}}} = 6.50 \text{ GHz}. \tag{18}$$

The average supply current obtained from the transient simulation was $286.3\ \mu$A. Consequently, the power consumption of the oscillator is

$$P_{\text{dc}} = V_{\text{DD}} I_{\text{avg}} = 1.0 \text{ V} \times 286.3\ \mu\text{A} = 286.3\ \mu\text{W}. \tag{19}$$

Using (8), the average propagation delay of each inverter stage is estimated to be

$$t_{\text{d,post-layout}} = \frac{T_{\text{post-layout}}}{2N} = 25.6 \text{ ps}. \tag{20}$$

B. Effect of Layout Parasitics

Table III compares the pre-layout and post-layout performance. The oscillation frequency decreases from 11.15 GHz to 6.50 GHz after parasitic extraction, corresponding to a frequency reduction of approximately 41.7%. In addition, the average stage delay increases from 14.95 ps to 25.6 ps, a 71.2% increase. Because the transistor dimensions (W/L) and bias conditions remained identical across both setups, this degradation is driven entirely by the distributed RC parasitics embedded in the DSPF netlist.

TABLE III
PRE-LAYOUT VERSUS POST-LAYOUT PERFORMANCE

Parameter	**Pre-Layout**	**Post-Layout**	**Change**
Oscillation frequency	11.15 GHz	6.50 GHz	-41.7%
Loop period	89.71 ps	153.84 ps	$+71.5\%$
Stage delay t_d	14.95 ps	25.6 ps	$+71.2\%$

These differences indicate that distributed interconnect parasitics have a significant influence on the delay characteristics of the oscillator. The results highlight the importance of post-layout extraction and verification for obtaining realistic performance estimates in nanoscale CMOS circuits.

C. Phase Noise and Figure-of-Merit Performance

Fig. 5 shows the simulated phase-noise spectrum of the oscillator. The curve exhibits a consistent $1/f^2$ slope across the offset spectrum, confirming that white thermal-noise contributions dominate the high-frequency phase-error profile [6], [9].

At an offset frequency of 1 MHz from the 6.50-GHz carrier, the single-sideband phase noise is

$$\mathcal{L}(1 \text{ MHz}) = -77.6 \text{ dBc/Hz}. \tag{21}$$

At an offset frequency of 100 MHz, the phase noise decreases to -124.3 dBc/Hz. Substituting the extracted post-layout metrics ($\mathcal{L}(1 \text{ MHz}) = -77.6$ dBc/Hz, $f_0 = 6.50$ GHz, and $P = 286.3\ \mu$W) into (16) yields

$$\begin{aligned} \text{FoM} &= -77.6 - 20\log_{10}\left(\frac{6.5 \times 10^9}{1.0 \times 10^6}\right) + 10\log_{10}(0.2863) \\ &= -77.6 - 76.26 - 5.43 = -159.3 \text{ dBc/Hz}. \end{aligned} \tag{22}$$

The relatively low power consumption of $286.3\ \mu$W contributes to the obtained FoM value. Although the phase noise is typical of a conventional single-ended ring oscillator, the results demonstrate that the proposed design achieves a reasonable trade-off among oscillation frequency, power consumption, and spectral purity.

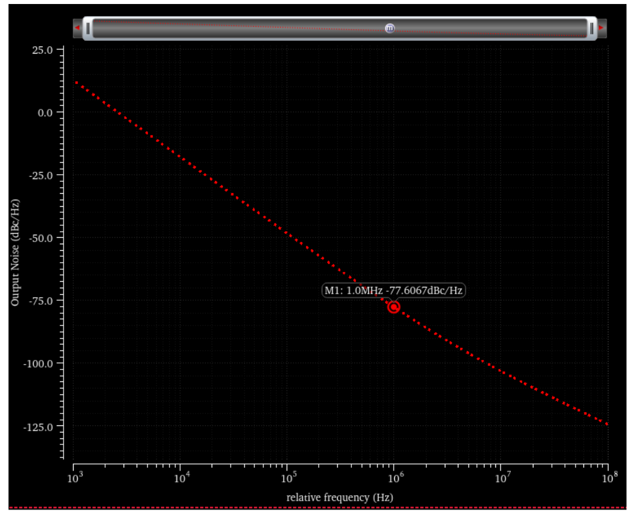

Fig. 5. Post-layout (RC-extracted) single-sideband phase noise versus relative offset frequency, showing -77.6 dBc/Hz at a 1 MHz offset.

D. Comparison with Previous Works

Table IV compares the proposed oscillator with representative CMOS ring oscillators reported in the literature. As the design is characterized by post-layout simulation, comparison with measured silicon should be treated with caution. Even so, the proposed oscillator attains the best figure of merit (-159.3 dBc/Hz), surpassing the current-controlled RO in [10] (-154.4 dBc/Hz), the switched-capacitor VCO in [12] (-155.5 dBc/Hz), and the current-starved VCO in [13] (-152.6 dBc/Hz). It also uses the lowest supply (1.0 V), the most compact core area ($73.96\ \mu\text{m}^2$), and sub-milliwatt power ($286.3\ \mu$W) on par with the lowest-power designs, while being the only one verified through a complete post-layout, parasitic-extracted flow. Other topologies, such as the quadrature design in [11], offer additional multiphase outputs at the cost of more delay stages.

TABLE IV
PERFORMANCE COMPARISON WITH REPORTED CMOS RING OSCILLATORS

Parameter	**This Work**	[10] (2012)	[12] (2017)	[13] (2023)
Technology	65 nm LP	0.18 μm	90 nm	90 nm
Topology	3-st. RO	3-st. CCO	SC-VCO	CS-VCO
Verification	Post-layout	Measured	Schematic	Sim.
Supply (V)	1.0	1.8	1.8	1.8
Freq. (GHz)	6.50	5.5	4.52–6.02	5.1
Power	286.3 μW	≤ 17.6 mW	0.295 mW	289 μW
Phase noise[a]	-77.6	-104[b]	N/R	-74
FoM (dBc/Hz)	$\mathbf{-159.3}$	-154.4	-155.5	-152.6
Area (μm^2)	73.96	300[c]	N/R	N/R

RO: ring oscillator; CCO: current-controlled oscillator; SC-VCO: switched-capacitor ring VCO; CS-VCO: current-starved VCO; Sim.: schematic-level simulation; N/R: not reported. [a]In dBc/Hz at 1 MHz offset unless noted. [b]Reported at 4 MHz offset. [c]Core area $0.0003\ \text{mm}^2$.

Compared with previous studies, the main contribution of this work lies not in introducing a new oscillator topology, but in demonstrating a comprehensive methodology for physical implementation, parasitic extraction, and post-layout characterization of a high-frequency ring oscillator in 65-nm CMOS technology.

VI. Conclusion

This paper presented a comprehensive parasitic-aware post-layout design and verification methodology for a low-power three-stage ring oscillator implemented in a 65-nm Low-Power CMOS process. The proposed approach covers the complete design flow, including custom layout generation, physical verification, RC parasitic extraction, and post-layout characterization using periodic steady-state (PSS) and periodic noise (`pnoise`) analyses.

The extracted oscillator achieved a stable operating frequency of 6.50 GHz while consuming only 286.3 μW from a 1.0-V supply. The phase-noise performance was characterized, resulting in a single-sideband phase noise of -77.6 dBc/Hz at a 1-MHz offset and a corresponding figure of merit of -159.3 dBc/Hz, occupying a compact core area of 73.96 μm^2.

The comparison between schematic-level and post-layout simulations demonstrates the importance of incorporating interconnect parasitics into the design flow. The results show that parasitic effects have a significant influence on the delay characteristics and operating frequency of nanoscale CMOS oscillators, highlighting the need for post-layout verification to obtain realistic performance estimates. Overall, the proposed methodology provides a practical framework for the design and evaluation of low-power, multi-gigahertz oscillator blocks in advanced CMOS technologies. Future work will focus on experimental silicon validation, the addition of explicit frequency-tuning capability, and the application of low-noise delay-cell techniques to further improve phase-noise performance.

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